Lab Experiment-15

Aim

To write a Python program to implement Probabilistic Context-Free Grammar (PCFG) parsing and find the most probable parse tree for a given sentence.

Algorithm

1. Import the required classes from the NLTK library.
2. Define a Probabilistic Context-Free Grammar (PCFG) with production rules and probabilities.
3. Create a Viterbi parser using the defined grammar.
4. Give a sentence as input.
5. Split the sentence into individual words.
6. Parse the sentence using the Viterbi parser.
7. Display the most probable parse tree and its probability.

Python Code

```
import nltk

from nltk import PCFG

from nltk.parse import ViterbiParser

grammar = PCFG.fromstring("""

S -> NP VP [1.0]

NP -> Det N [0.8] | 'John' [0.2]

VP -> V NP [1.0]

Det -> 'the' [0.6] | 'a' [0.4]

N -> 'cat' [0.5] | 'dog' [0.5]
```

```
V -> 'sees' [1.0]

""")

parser = ViterbiParser(grammar)

sentence = "John sees the cat"

tokens = sentence.split()

for tree in parser.parse(tokens):

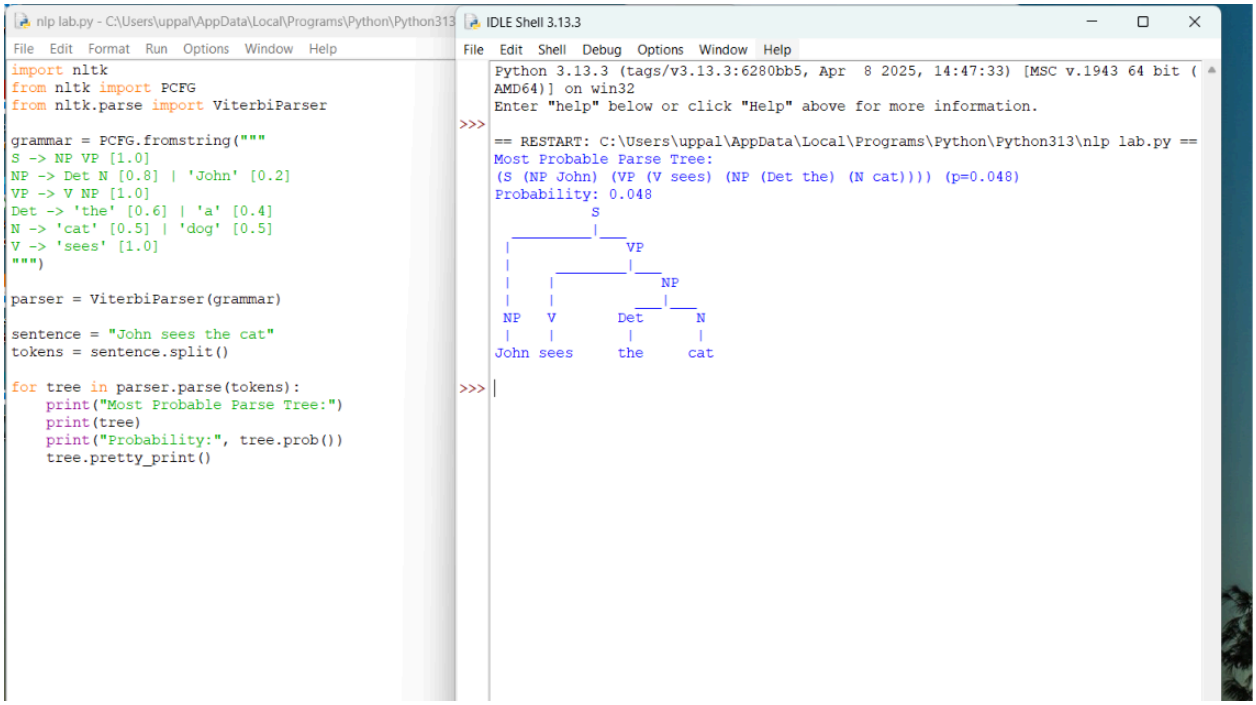
    print("Most Probable Parse Tree:")

    print(tree)

    print("Probability:", tree.prob())

    tree.pretty_print()
```

Output:



Result

Thus, the Python program successfully implements PCFG parsing and generates the most probable parse tree for the given sentence.